TUTORIAL 8

Linear Maps and Coordinates

MOTIVATION AND GUIDING IDEA

Motivation. The same information may appear as a polynomial, a matrix or a coordinate vector. A linear map compares such descriptions while preserving linear combinations. The kernel asks what is lost; the image asks what can be attained.

NOTATION AND CONVENTIONS

- $T : V \to W$ denotes a map with domain V and codomain W; $T(v) \in W$ is the image of v. The vectors $0_V, 0_W$ are the zero vectors in the corresponding spaces.
- $\ker T = \{v \in V : T(v) = 0_W\}$ is the kernel; $\operatorname{Im} T = T(V) = \{T(v) : v \in V\}$ is the image. Do not automatically identify the image with the codomain.
- $\operatorname{rank} T = \dim \operatorname{Im} T$ is the rank; $\dim \ker T$ is the nullity. The notation $V \cong W$ means that a linear isomorphism between them exists.
- $B = (v_1, \ldots, v_n)$ is an **ordered** basis. If $v = \sum_{j=1}^{n} a_j v_j$, then $[v]_B = (a_1, \ldots, a_n)^T$. The order of basis vectors determines the order of entries.
- $\mathbb{F}_{\leq n}[X]$ is the space of polynomials of degree at most n, including the zero polynomial. For $T : V \to V$, $T^2 = T \circ T$ denotes composition, not entrywise squaring.

DEFINITION 8.1

Let V, W be vector spaces over the same field $\mathbb{F}$. A map $T : V \to W$ is linear if it is well defined and satisfies

$$T(av + bw) = aT(v) + bT(w) \qquad (v, w \in V,\ a, b \in \mathbb{F}).$$

It is injective if $T(v) = T(w)$ implies $v = w$, and surjective if every $y \in W$ equals $T(v)$ for some $v \in V$. For a linear map, injectivity is equivalent to $\ker T = \{0_V\}$, and surjectivity to $\operatorname{Im} T = W$. An isomorphism is a bijective linear map; its inverse is linear. If $\dim V < \infty$, then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

The coordinate map $v \mapsto [v]_B$ is an isomorphism $V \to \mathbb{F}^n$. A vector and its coordinate column are related by a choice of basis; they should not be equated without an appropriate identification.

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MOTIVATION AND GUIDING IDEA

Tutorial objectives. Recognize linear maps, compute kernels and images, and use coordinates to pass between polynomials, matrices and vectors. Distinguish a mathematical object from its representation in a chosen basis.

8.1 Linear maps

DEFINITION 8.2

Let V, W be vector spaces over the same field $\mathbb{F}$.
A map $T : V \to W$ is **linear** if, for all $v_1, v_2 \in V$ and $\lambda \in \mathbb{F}$,

1. $T(v_1 + v_2) = T(v_1) + T(v_2)$.
2. $T(\lambda v_1) = \lambda T(v_1)$.

Interpretation A linear map sends vectors from one vector space to another while preserving their linear structure. It respects vector addition and scalar multiplication.

The first condition says that adding and then applying the map gives the same result as applying the map and then adding. Thus addition commutes with applying the map.

The second condition says that multiplying by a scalar before applying the map gives the same result as multiplying the image by that scalar. In a real geometric setting, scaling an input by λ scales its image by the same factor.

NOTE

Together these properties imply that, for all $\alpha, \beta \in \mathbb{F}$ and $v_1, v_2 \in V$,

$$T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2),$$

so a linear map preserves every linear combination.
Geometrically, real linear maps include scalings, rotations and reflections. They send a line to a line or a point and fix the zero vector. In particular, $T(0_V) = 0_W$ always holds.
The main consequences are:

- More generally, $T(\sum_{i=1}^{n} \lambda_i v_i) = \sum_{i=1}^{n} \lambda_i T(v_i)$.
- A necessary condition for $T : V \to W$ to be linear is $T(0_V) = 0_W$. (A map with $T(0_V) \neq 0_W$ is not linear.)

EXERCISE 8.1

$$V = \mathbb{R}_{\leq n}[X], \qquad n \geq 1.$$
$$T_1(P) = P(X) + 2P(-1), \qquad T_2(P) = X^2 P(X),$$
$$T_3(P) = P(X + 2) - P(2).$$

Please decide which formulas define linear maps $V \to V$.

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EXERCISE 8.1 CONTINUED

WORKED SOLUTION

T_1 and T_3 do not increase polynomial degree, so they are well-defined and linear. The second formula does not define a self-map of V, since it sends X^n outside V.

$$T_1(\alpha P + \beta Q) = \alpha P + \beta Q + 2\alpha P(-1) + 2\beta Q(-1) = \alpha T_1(P) + \beta T_1(Q).$$

$$T_2(X^n) = X^{n+2} \notin V.$$

$$T_3(\alpha P + \beta Q) = \alpha P(X+2) + \beta Q(X+2) - \alpha P(2) - \beta Q(2) = \alpha T_3(P) + \beta T_3(Q).$$

We've got it. Victory!

MOTIVATION AND GUIDING IDEA

Two guiding questions. The kernel records information lost by a map; the image records attainable outputs. A trivial kernel means that distinct inputs remain distinguishable. An image equal to the codomain means that every output in that codomain is attainable.

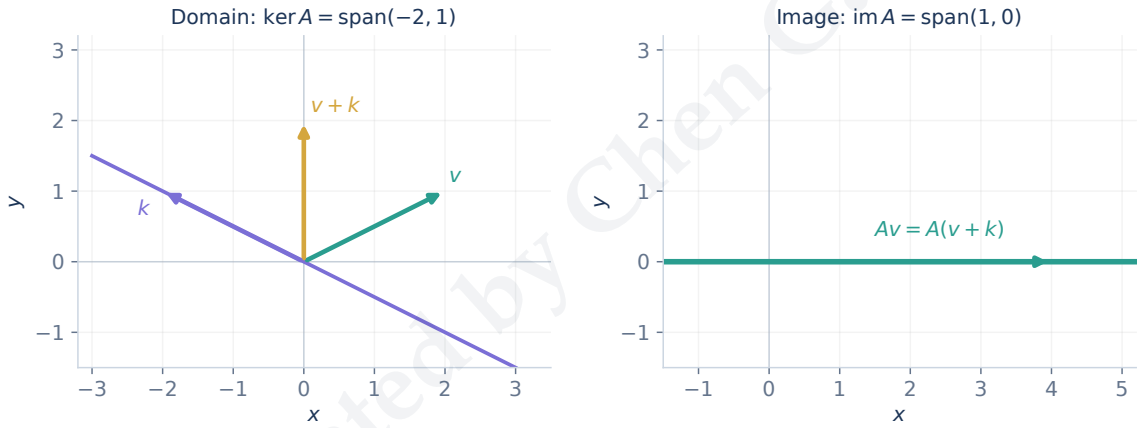


Figure 8.1. The kernel lies in the domain and the image in the codomain. Several inputs can have the same output when the kernel is nontrivial.

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8.2 The kernel and image of a linear map

RECALL AND TOOLS

Let $T : V \to W$ be linear.
The **kernel of** T consists of all vectors sent to 0_W:

$$\ker(T) = \{v \in V | T(v) = 0\}$$

The kernel is a subspace of V.
The **image of** T is the set of all outputs attained by the map:

$$\mathrm{Im}(T) = \{T(v) | v \in V\} = \{w \in W | \exists v \in V, T(v) = w\}$$

The image is a subspace of W. It consists of outputs actually attained, whereas the codomain W is specified as part of the definition of the map. These need not be equal.

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The sets $\ker T$ and $\operatorname{Im} T$ are nonempty, since $T(0_V) = 0_W$.

- Always $\operatorname{Im} T \subseteq W$. If $\operatorname{Im} T = W$, the map is **surjective**, or **onto**.
- T is **injective**, or **one-to-one**, if, for all $v_1, v_2 \in V$,

$$T(v_1) = T(v_2) \Longrightarrow v_1 = v_2$$

  Equivalently, no two distinct inputs have the same image.

$$v_1 \neq v_2 \Longrightarrow T(v_1) \neq T(v_2)$$

- A linear map T is an **isomorphism** if it is injective and surjective.
- Vector spaces V, W are **isomorphic** if there **exists** an isomorphism $T : V \to W$. We then write $V \cong W$.

PROPOSITION 8.1

A linear map $T : V \to W$ is **injective** if and only if $\ker T = \{0_V\}$.
For a homogeneous linear system, this means that the trivial solution is the only solution.

PROPOSITION 8.2

Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

1. If T is injective, $\dim V \leq \dim W$.
2. If T is surjective, $\dim W \leq \dim V$.
3. If $\dim V = \dim W$, then T is injective **if and only if** it is surjective.

EXERCISE 8.2

Let $n \geq 2$ and $V = M_n(\mathbb{F})$. Define $T : V \to \mathbb{F}$ by

$$T(A) = \operatorname{Tr}(A).$$

1. Please prove that T is linear.
2. Please find $\ker T$.
3. Please find $\operatorname{Im} T$.
4. Please compute $\dim \ker T$ and $\dim \operatorname{Im} T$, and please verify rank-nullity.

WORKED SOLUTION

Write $A = (a_{ij}), B = (b_{ij})$.

1. Linearity:

$$T(A+B) = \operatorname{Tr}(A+B) = \sum_{i=1}^{n}(a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} = T(A) + T(B),$$

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EXERCISE 8.2 CONTINUED

For every $\lambda \in \mathbb{F}$,

$$T(\lambda A) = \mathrm{Tr}(\lambda A) = \sum_{i=1}^{n} \lambda a_{ii} = \lambda \sum_{i=1}^{n} a_{ii} = \lambda T(A).$$

Thus T is linear.

2. Kernel:
$$\ker(T) = \{A \in M_n(\mathbb{F}) \ : \ \mathrm{Tr}(A) = 0\}.$$
This is the subspace of matrices with trace zero.

3. Image: for any $\alpha \in \mathbb{F}$, take
$$A = \mathrm{diag}(\alpha, 0, \ldots, 0) \in M_n(\mathbb{F}),$$
Then $\mathrm{Tr}(A) = \alpha$, so
$$\mathrm{Im}(T) = \mathbb{F}.$$

4. Dimensions: $\dim M_n(\mathbb{F}) = n^2$. Since $\mathrm{Im}\, T = \mathbb{F}$, its dimension is 1. Rank-nullity therefore gives
$$\dim \ker(T) = n^2 - 1.$$
Alternatively, $n^2 - 1$ entries may be chosen freely, and the last diagonal entry is fixed by requiring trace zero.

That does it. Victory!

EXERCISE 8.3

$$V = C[-1, 2], \qquad S : V \to \mathbb{R}, \qquad S(f) = \int_{-1}^{2} f(x)\, dx.$$

Please prove that S is linear. Please find its kernel and image and please give a nonzero kernel element.

WORKED SOLUTION

Linearity of integration proves linearity. Every real value is obtained from a suitable constant function. The function $x - \frac{1}{2}$ is nonzero and lies in the kernel.

$$S(\alpha f + \beta g) = \int_{-1}^{2} (\alpha f + \beta g) = \alpha \int_{-1}^{2} f + \beta \int_{-1}^{2} g = \alpha S(f) + \beta S(g).$$

$$\ker S = \left\{ f \in C[-1, 2] : \int_{-1}^{2} f(x)\, dx = 0 \right\}, \qquad \mathrm{Im}\, S = \mathbb{R}.$$

$$S(c/3) = \frac{c}{3}(2 - (-1)) = c, \qquad S(x - \tfrac{1}{2}) = \left[\tfrac{x^2}{2} - \tfrac{x}{2} \right]_{-1}^{2} = 1 - 1 = 0.$$

All wrapped up. Victory!

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EXERCISE 8.4

Let V be a vector space over a field F, and let $T : V \to V$ be linear. Please prove:
(a) $\ker T \subseteq \ker T^2$; (b) $\operatorname{Im} T^2 \subseteq \operatorname{Im} T$.

WORKED SOLUTION

(a) We must show that every vector in the set on the left also belongs to the set on the right.
Let $v \in \ker T$. Then $T^2(v) = T(T(v)) = T(0) = 0$, so $v \in \ker T^2$.
(b) Let $w \in \operatorname{Im} T^2$. Some $v \in V$ satisfies $w = T^2(v) = T(T(v))$.
Set $u = T(v)$. Then $w = T(u)$, so $w \in \operatorname{Im} T$.

We've reached the finish. Victory!

DEFINITION 8.3

If $T : V \to W$ is an **isomorphism**, each $w \in W$ has a unique $v \in V$ with $T(v) = w$.
Thus the **inverse map** $T^{-1} : W \to V$ is defined by $T^{-1}(w) = v$ whenever $T(v) = w$.
Applying T to v gives w; applying T^{-1} to w recovers v.

EXERCISE 8.5

$$V = M_2(\mathbb{R}), \qquad \Phi\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a, b, c, d)^T.$$

$$U = \{x \in \mathbb{R}^4 : x_2 = x_3\}, \qquad W = \{x \in \mathbb{R}^4 : x_1 = x_4 = 0,\ x_2 = -x_3\}.$$

Please prove that Φ is an isomorphism. Please find $\Phi^{-1}(U)$ and $\Phi^{-1}(W)$ and their dimensions.

WORKED SOLUTION

Listing matrix entries preserves addition and scalar multiplication; the inverse formula proves bijectivity. The preimages are the symmetric and skew-symmetric matrices. Their displayed free parameters give the independent spanning matrices and the dimensions.

$$\Phi^{-1}((a, b, c, d)^T) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

$$\Phi^{-1}(U) = \left\{ \begin{pmatrix} a & b \\ b & d \end{pmatrix} : a, b, d \in \mathbb{R} \right\} = \operatorname{span}\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

$$\Phi^{-1}(W) = \left\{ \begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix} : b \in \mathbb{R} \right\} = \operatorname{span}\left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}.$$

$$\dim U = 3, \qquad \dim W = 1.$$

Yes! Victory!

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8.3 Rank-nullity for linear maps

PROPOSITION 8.3

Let $T : V \to W$ be linear, with V finite-dimensional.
Then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

EXERCISE 8.6

(a) $T : \mathbb{R}^3 \to \mathbb{R}^3, \quad \ker T = \operatorname{span}\{(1,-1,2)^T\}, \quad \operatorname{Im} T = \operatorname{span}\{(2,0,1)^T\};$

(b) $T : M_2(\mathbb{R}) \to \mathbb{R}^2, \quad \ker T = \operatorname{span}\left\{\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}\right\}.$

Please decide whether a linear map exists in each case.

WORKED SOLUTION

Neither map exists. Part (a) contradicts rank-nullity. In part (b), the image would have to be a three-dimensional subspace of a two-dimensional space.

(a) $\dim \ker T + \dim \operatorname{Im} T = 1 + 1 = 2 \neq 3.$

(b) $\dim \operatorname{Im} T = \dim M_2(\mathbb{R}) - \dim \ker T = 4 - 1 = 3 > 2.$

V-I-C-T-O-R-Y!

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8.4 Coordinates

DEFINITION 8.4

Let V be a vector space over a field $\mathbb{F}$ with ordered basis $B = (v_1, \ldots, v_n)$.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors. Thus there are unique $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$ such that

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n$$

The coefficients $\alpha_1, \ldots, \alpha_n$ are the **coordinates of** v **relative to** B.

The column vector $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{F}^n$ is the **coordinate vector of** v **relative to** B, denoted $[v]_B$.

EXERCISE 8.7

$$B = ((1,1,0)^T, (0,1,1)^T, (1,0,1)^T), \qquad v = (5,3,6)^T.$$

Please find $[v]_B$ and please check it by reconstructing v.

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EXERCISE 8.7 CONTINUED

WORKED SOLUTION

The basis-column matrix has determinant 2, so it is a basis. Equating the components of the linear combination gives the three equations.

$$a(1,1,0)^T + b(0,1,1)^T + c(1,0,1)^T = (a+c, a+b, b+c)^T.$$

$$a+c=5, \quad a+b=3, \quad b+c=6 \implies a=1, \quad b=2, \quad c=4.$$

$$[v]_B = (1,2,4)^T, \qquad (1,1,0)^T + 2(0,1,1)^T + 4(1,0,1)^T = (5,3,6)^T.$$

Victory!

Conversely, knowing a vector's coordinates in a chosen basis allows us to reconstruct the vector.

With the notation above, $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ **if and only if** $[v]_B = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$.

EXERCISE 8.8

$$B = ((2,1)^T, (-1,1)^T), \qquad [v]_B = (3,-2)^T.$$

Please find v in standard coordinates.

WORKED SOLUTION

The coordinate entries are the coefficients of the ordered basis vectors.

$$v = 3(2,1)^T - 2(-1,1)^T = (6,3)^T + (2,-2)^T = (8,1)^T.$$

And we're done. Victory!

PROPOSITION 8.4

Let V have dimension n over a field $\mathbb{F}$, and let B be an ordered basis of V.
The coordinate map $T_B : V \to \mathbb{F}^n$, defined by $T_B(v) = [v]_B$, is an isomorphism. In particular, $V \cong \mathbb{F}^n$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F}$, let B be an ordered basis with n vectors, and let $u_1, \ldots, u_m$ be vectors in V. Then:

1. $\{u_1, \ldots, u_m\}$ is linearly independent (in V) **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ is linearly independent (in $\mathbb{F}^n$).

2. $\{u_1, \ldots, u_m\}$ spans V **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ spans $\mathbb{F}^n$.

3. $\{u_1, \ldots, u_m\}$ is a basis of V **if and only if** $\{[u_1]_B, \ldots, [u_m]_B\}$ is a basis of $\mathbb{F}^n$.

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EXERCISE 8.9

$$p_1 = 1 + 2X, \qquad p_2 = X + 3X^2, \qquad p_3 = 1 + X + 2X^2.$$

Please prove that these polynomials are independent in $\mathbb{R}_2[X]$ using coordinates in $(1, X, X^2)$.

WORKED SOLUTION

The coordinate columns are therefore independent. The coordinate map is an isomorphism, so the polynomials are independent as well.

$$[p_1] = (1, 2, 0)^T, \quad [p_2] = (0, 1, 3)^T, \quad [p_3] = (1, 1, 2)^T.$$

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 3 & 2 \end{pmatrix} = (2 - 3) + (6 - 0) = 5 \neq 0.$$

You know what that means? V-I-C-T-O-R-Y!

EXERCISE 8.10

$$U = \operatorname{span}\{U_1, U_2, U_3\}, \qquad V = \operatorname{span}\{V_1, V_2\} \subseteq M_2(\mathbb{R}),$$

$$U_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}, \quad V_1 = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Please find bases and dimensions for U, V, and $U \cap V$.

WORKED SOLUTION

In each list, the entries uniquely determine the coefficients, proving independence. Membership in U requires a relation between the second-row entries. Substituting a general element of V determines the intersection.

$$U = \left\{ \begin{pmatrix} a & b \\ c & 2c \end{pmatrix} : a, b, c \in \mathbb{R} \right\}, \qquad V = \left\{ \begin{pmatrix} r & s \\ r + s & 2r \end{pmatrix} : r, s \in \mathbb{R} \right\}.$$

$$\begin{pmatrix} r & s \\ r + s & 2r \end{pmatrix} \in U \iff 2r = 2(r + s) \iff s = 0.$$

$$B_U = (U_1, U_2, U_3), \quad \dim U = 3, \qquad B_V = (V_1, V_2), \quad \dim V = 2,$$

$$U \cap V = \operatorname{span}\{V_1\}, \qquad \dim(U \cap V) = 1.$$

Another one down. Victory!

8.5 Five descriptions of a linear map

Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ so that, for every

$$(x, y) \in \mathbb{R}^2$$

we have

$$T(x, y) = (2x + 2y,\ 2x - 2y).$$

We now describe this same map in several ways.

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Method 1: A formula for a general vector

This is a **general formula**: it specifies $T(v)$ for arbitrary $v = (x, y) \in \mathbb{R}^2$, rather than just particular vectors:

$$T(x, y) = (2x + 2y,\ 2x - 2y).$$

It defines the map directly on its entire domain.

Method 2: Images of a basis

Choose the standard basis:

$$e_1 = (1, 0), \qquad e_2 = (0, 1).$$

Then

$$T(e_1) = T(1, 0) = (2, 2), \qquad T(e_2) = T(0, 1) = (2, -2).$$

For $(x, y) = xe_1 + ye_2$, extend linearly:

$$T(x, y) = xT(e_1) + yT(e_2) = x(2, 2) + y(2, -2) = (2x + 2y,\ 2x - 2y).$$

Thus specifying the images of a basis defines the same map.

Method 3: A matrix

Write the input as a column:

$$\begin{pmatrix} x \\ y \end{pmatrix}.$$

Define

$$A = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}.$$

Then

$$T(x, y) = A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x + 2y \\ 2x - 2y \end{pmatrix}.$$

This matrix represents the same map in the standard basis.

Method 4: Familiar linear operations

View T as performing the following operations:

- First component: twice the sum of the coordinates.
- Second component: twice the difference of the coordinates.

Thus

$$T(x, y) = \big(2\text{sum}(x, y), 2\text{diff}(x, y)\big),$$

These operations are linear, so T is linear.

Method 5: A structural decomposition

Specify that T satisfies

$$T(e_1 + e_2) = (4, 0), \qquad T(e_1 - e_2) = (0, 4),$$

and that T is linear.

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Since $\{e_1+e_2, e_1-e_2\}$ is a basis of $\mathbb{R}^2$, these requirements determine T uniquely. Calculation gives the same formula:

$$T(x,y) = (2x+2y,\ 2x-2y).$$

Remarks

- A **general formula** defines $T(v)$ for an arbitrary vector v.
- A linear map may be specified by a formula, basis images, a matrix or a structural description.
- The descriptions differ, but the map itself is the same.

8.5.1 Summary of the methods

RECALL AND TOOLS

1. **An explicit formula for a general vector**
   Specify $T(v)$ for every $v \in V$.
   Check: the formula is well defined and satisfies both linearity conditions.

2. **Basis images and linear extension**
   If $B = \{v_1, \dots, v_n\}$ is a basis of V, freely choose

   $$T(v_1), \dots, T(v_n) \in W$$

   and define, for $v = \sum_{i=1}^{n} \alpha_i v_i$,

   $$T(v) := \sum_{i=1}^{n} \alpha_i\, T(v_i).$$

   Each choice of basis images determines exactly one linear map.

3. **A matrix**
   For $V = \mathbb{F}^n$ and $W = \mathbb{F}^m$, define

   $$T(x) = Ax$$

   where A is an $m \times n$ matrix.
   Every matrix defines a linear map. After choosing ordered bases, every linear map between finite-dimensional spaces has a matrix representation.
   Remark: the matrix depends on the chosen bases; the map itself does not.

4. **A familiar linear operation**
   Use an operation such as differentiation, integration or multiplication by a fixed scalar.
   Examples:

   $$T(P) = P', \qquad T(f) = \int_{-1}^{2} f(x)\,dx.$$

   Specify a domain and codomain, check well-definedness and verify linearity in each case.

5. **A direct-sum decomposition**
   A map can be specified by its action on the components of a direct sum.
   For example, if $V = U \oplus W$, define T by

   $$T(u) = 0 \ \ (\forall u \in U), \qquad T(w) = w \ \ (\forall w \in W).$$

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Linearity follows from the direct-sum decomposition and the linear action on each component.

Common errors

- Adding a nonzero constant vector destroys linearity: $T(v) = v + b$ with $b \neq 0$ is not linear.
- Always specify the domain and codomain: $T : V \to W$. The image $\operatorname{Im} T$ may be a proper subspace of the codomain.
- Distinguish a vector, its coordinate vector in an ordered basis, and the matrix representing a map.

EXERCISE 8.11

$$P : \mathbb{R}^4 \to \mathbb{R}^4, \qquad P(x) = x - \frac{x_1 + x_2 + x_3 + x_4}{4}(1,1,1,1)^T.$$

Please prove that P is linear, please find its kernel and image, and please prove $P^2 = P$.

WORKED SOLUTION

The image lies in the zero-sum subspace. Every vector in that subspace is fixed by P, proving equality with the image. The kernel and image dimensions are $1, 3$, as rank-nullity requires.

$$\mu(x) = \frac{x_1 + x_2 + x_3 + x_4}{4},$$

$$\begin{aligned} P(\alpha x + \beta y) &= \alpha x + \beta y - (\alpha\mu(x) + \beta\mu(y))(1,1,1,1)^T \\ &= \alpha P(x) + \beta P(y). \end{aligned}$$

$$P(x) = 0 \iff x = \mu(x)(1,1,1,1)^T, \qquad \ker P = \operatorname{span}\{(1,1,1,1)^T\}.$$

$$\sum_{j=1}^{4} (Px)_j = \sum_{j=1}^{4} x_j - 4\mu(x) = 0.$$

$$\operatorname{Im} P = \{y \in \mathbb{R}^4 : y_1 + y_2 + y_3 + y_4 = 0\} = \operatorname{span}\{e_1 - e_4, e_2 - e_4, e_3 - e_4\}.$$

$$\mu(Px) = 0 \quad \Longrightarrow \quad P(Px) = Px.$$

We've got it. Victory!

Additional practice

EXERCISE 8.12 FOUNDATION

$$T : \mathbb{R}_{\leq 2}[X] \to \mathbb{R}^2, \qquad T(p) = (p(2), p(-2))^T.$$

Please prove linearity, please find the kernel and image, and please check rank-nullity. Please construct a linear map S in the opposite direction with $T \circ S = I$. Please decide whether $S \circ T = I$.

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EXERCISE 8.12 FOUNDATION CONTINUED

WORKED SOLUTION

Evaluation preserves addition and scalar multiplication. The formula for S is linear and takes the required values at the two points, so T is onto. The reverse composition preserves the evaluations but loses the quadratic coefficient, so it is not the identity.

$$p = a + bX + cX^2 \in \ker T \iff a + 2b + 4c = 0, \quad a - 2b + 4c = 0 \iff b = 0,\ a = -4c.$$

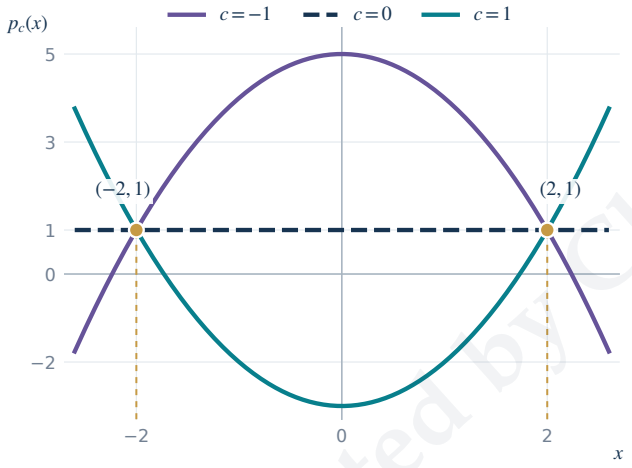
$$\ker T = \operatorname{span}\{X^2 - 4\}, \quad \dim \ker T = 1, \qquad S((u,v)^T) = \frac{u+v}{2} + \frac{u-v}{4}X.$$

$$(T \circ S)(u,v)^T = (u,v)^T, \qquad \operatorname{Im} T = \mathbb{R}^2, \quad 1 + 2 = 3.$$

$$(S \circ T)(a + bX + cX^2) = a + 4c + bX, \qquad (S \circ T)(X^2) = 4 \neq X^2.$$

That does it. Victory!

$$p_c(x) = 1 + c(x^2 - 4), \qquad c \in \{-1, 0, 1\}$$



$$T(p) = (p(2), p(-2))^T$$

$$p_{-1}(x) = 5 - x^2, \quad p_0(x) = 1, \quad p_1(x) = x^2 - 3$$

$$p_c(-2) = p_c(2) = 1$$

$$T(p_{-1}) = T(p_0) = T(p_1) = (1,1)^T$$

$$\ker T = \operatorname{span}\{X^2 - 4\}$$

$$p_c - p_0 = c(X^2 - 4)$$

Figure 8.2. Three distinct polynomials give the same two measurements: $p_c(2) = p_c(-2) = 1$ for $c = -1, 0, 1$. Their differences $p_c - p_0 = c(X^2 - 4)$ lie in the kernel of $T(p) = (p(2), p(-2))^T$, so these measurements do not distinguish the polynomials.

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EXERCISE 8.13 CHALLENGE

$$P : V \to V, \quad P^2 = P.$$

$$V = \ker P \oplus \operatorname{Im} P, \quad \ker(I - P) = \operatorname{Im} P, \quad \operatorname{Im}(I - P) = \ker P.$$

$$P_0(x,y,z)^T = (x + 2z, y - z, 0)^T \quad \text{on } \mathbb{R}^3.$$

Please prove the three identities and the existence and uniqueness of the projection with kernel U and image W when $V = U \oplus W$. Please compute the explicit decomposition for P_0.

WORKED SOLUTION

The first decomposition splits a vector into a kernel part and an image part. The map acts as the identity on its image, so the intersection is zero. The displayed calculations prove all three identities. For a given $U \oplus W$, uniqueness of decomposition makes $P(u + w) = w$ well-defined and linear. It is the only possibility because it must vanish on the kernel and fix the

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EXERCISE 8.13 CHALLENGE CONTINUED

image. In the example, the two parts lie in the stated subspaces.

$$v = (v - Pv) + Pv, \quad P(v - Pv) = 0, \quad P(Pv) = Pv.$$

$$w \in \ker P \cap \operatorname{Im} P \implies Pw = 0 = w \implies w = 0.$$

$$(I - P)v = 0 \iff Pv = v \iff v \in \operatorname{Im} P, \quad P(I - P) = 0, \quad u \in \ker P \implies (I - P)u = u.$$

$$P(u + w) = w \quad (u \in U, w \in W),$$

$$\ker P_0 = \operatorname{span}\{(-2, 1, 1)^T\}, \quad \operatorname{Im} P_0 = \{(a, b, 0)^T : a, b \in \mathbb{R}\},$$

$$(x, y, z)^T = z(-2, 1, 1)^T + (x + 2z, y - z, 0)^T.$$

All wrapped up. Victory!

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EXERCISE 8.14 CHALLENGE

Let V have finite dimension $n \geq 2$ over a field F. Let $T : V \to V$ be linear and satisfy $T(v) \in \text{span}\{v\}$ for every $v \in V$. Please prove that one scalar λ satisfies $T(v) = \lambda v$ for all v. Please check whether imposing the condition only on a fixed basis is sufficient.

WORKED SOLUTION

Choose a basis $e_1, \ldots, e_n$ and write $T(e_j) = \lambda_j e_j$. For $i \neq j$, the hypothesis for $e_i + e_j$ gives

$$\lambda_i e_i + \lambda_j e_j = T(e_i + e_j) = \mu(e_i + e_j)$$

for some μ. Independence implies $\lambda_i = \lambda_j = \mu$. All basis scalars therefore equal one λ, and linearity gives

$$T\left(\sum_j a_j e_j\right) = \sum_j a_j \lambda e_j = \lambda \sum_j a_j e_j.$$

A basis condition alone is insufficient. On $\mathbb{R}^2$, $T(x, y) = (x, 2y)$ preserves the lines of the two standard basis vectors, but $T(1, 1) = (1, 2)$ is not a multiple of $(1, 1)$.

We've reached the finish. Victory!

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EXERCISE 8.15 CHALLENGE

Let V, W, U be finite-dimensional spaces over a field F, and let $S : V \to W$ and $T : V \to U$ be linear. Please prove that $\ker S \subseteq \ker T$ if and only if there is a linear map $R : W \to U$ with $T = R \circ S$. When S is onto, please prove that R is unique.

WORKED SOLUTION

If $T = R \circ S$ and $x \in \ker S$, then $T(x) = R(0) = 0$, so $x \in \ker T$. Conversely, assume the kernel inclusion and define on $\operatorname{Im} S$

$$R_0(Sx) = Tx.$$

This is well-defined: if $Sx = Sy$, then $x - y \in \ker S \subseteq \ker T$, so $Tx = Ty$. Also

$$R_0(aSx + bSy) = R_0(S(ax + by)) = T(ax + by) = aTx + bTy,$$

which proves linearity. Choose a basis of $\operatorname{Im} S$ and extend it to a basis of W. Define R by R_0 on the first basis vectors and by zero on the added vectors, then extend linearly. This gives $R(Sx) = Tx$ for every x. If S is onto, every $w \in W$ is Sx for some x. The equality forces $R(w) = Tx$, so there is only one possible map R.

Yes! Victory!

DEFINITION 8.5

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m \times n}$. Write its columns as

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix}, \qquad a_j \in \mathbb{F}^m.$$

Let $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{F}^n$.

1. **A linear combination of columns.** Matrix-vector multiplication is defined by

   $$Av = v_1 a_1 + \cdots + v_n a_n.$$

   Thus Av is the linear combination of the columns of A with coefficients $v_1, \ldots, v_n$. It follows that

   $$\{Av \mid v \in \mathbb{F}^n\} = \operatorname{span}\{a_1, \ldots, a_n\} = \operatorname{Col}(A).$$

2. **A linear map.** The function

   $$T_A : \mathbb{F}^n \to \mathbb{F}^m, \qquad T_A(v) = Av,$$

   is linear: for all $u, v \in \mathbb{F}^n$ and $\lambda \in \mathbb{F}$,

   $$T_A(u + v) = A(u + v) = Au + Av,$$

   $$T_A(\lambda v) = A(\lambda v) = \lambda(Av).$$

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Thus multiplication by A is a linear map, and its columns describe its action on the standard basis of $\mathbb{F}^n$:

$$T_A(e_j) = a_j.$$

NOTATION AND CONVENTIONS

Notation guide. For an ordered basis $B = (b_1, \dots, b_n)$, the notation $[v]_B = (c_1, \dots, c_n)^T$ means $v = \sum_{j=1}^{n} c_j b_j$. A^{-1} denotes the inverse, A^T the transpose, and $(A \mid I_n)$ an augmented matrix. $R_i \leftrightarrow R_j$ interchanges rows; $R_i \leftarrow \lambda R_i$ requires $\lambda \neq 0$; $R_i \leftarrow R_i + \lambda R_j$ requires $i \neq j$. rank A is the number of pivots in a row-echelon form. Distinguish a matrix unit E_{ij} from an entire elementary matrix E.

EXERCISE 8.16 FOUNDATION

Let $w_1 = (1, 2, 0)^T$, $w_2 = (0, 1, -1)^T$, $e_3 = (0, 0, 1)^T$. Please prove that $B = (w_1, w_2, e_3)$ is a basis of $\mathbb{R}^3$. Please find the inverse of its column matrix by row reduction and please compute $[b]_B$ for $b = (3, 8, 5)^T$.

WORKED SOLUTION

The equation $aw_1 + bw_2 + ce_3 = 0$ gives $a = 0$, $2a + b = 0$, and $-b + c = 0$, so $a = b = c = 0$. Three independent vectors in $\mathbb{R}^3$ form a basis.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 \end{array}\right).$$

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 1 & 1 \end{pmatrix}, \qquad [b]_B = A^{-1} \begin{pmatrix} 3 \\ 8 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -6+8 \\ -6+8+5 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 7 \end{pmatrix}.$$

Indeed, $3w_1 + 2w_2 + 7e_3 = (3, 6 + 2, -2 + 7)^T = (3, 8, 5)^T$. The representation is unique because B is a basis.

V-I-C-T-O-R-Y!

EXERCISE 8.17

Please determine whether $B = (x^2 - x,\ x + 2,\ 1)$ is a basis of $\mathbb{R}_2[x]$ and of $\mathbb{R}_3[x]$. In the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

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EXERCISE 8.17 CONTINUED

WORKED SOLUTION

Write p as a combination with coefficients α, β, γ and compare coefficients:

$$ax^2 + bx + c = \alpha(x^2 - x) + \beta(x + 2) + \gamma.$$

$$\alpha = a, \quad -\alpha + \beta = b, \quad 2\beta + \gamma = c.$$

$$[p]_B = (a,\ a + b,\ c - 2a - 2b)^T.$$

For each a, b, c there is a unique coordinate triple, so B is a basis of $\mathbb{R}_2[x]$. Every combination has degree at most 2 and cannot equal x^3, so this is not a basis of $\mathbb{R}_3[x]$.

Victory!

EXERCISE 8.18

Please determine whether $B = (x^2 + 2x,\ x - 3,\ 2)$ is a basis of $\mathbb{R}_2[x]$ and of $\mathbb{R}_3[x]$. In the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Write p as a combination with coefficients α, β, γ and compare coefficients:

$$ax^2 + bx + c = \alpha(x^2 + 2x) + \beta(x - 3) + 2\gamma.$$

$$\alpha = a, \quad 2\alpha + \beta = b, \quad -3\beta + 2\gamma = c.$$

$$[p]_B = (a,\ b - 2a,\ (c + 3b - 6a)/2)^T.$$

For each a, b, c there is a unique coordinate triple, so B is a basis of $\mathbb{R}_2[x]$. Every combination has degree at most 2 and cannot equal x^3, so this is not a basis of $\mathbb{R}_3[x]$.

And we're done. Victory!

EXERCISE 8.19 EXAM QUESTION

1. Let $n \geq 2$ and $A, B \in M_n(\mathbb{R})$ satisfy $B \neq 0$ and $\operatorname{rank}(A) = n - 1$. Please prove that $AB = 0$ implies $\operatorname{rank}(B) = 1$.
2. Let $A, B \in M_{2031}(\mathbb{R})$ satisfy $AB = 0$. Please prove that
$$\operatorname{rank}(A) + \operatorname{rank}(B) \leq 2031.$$

WORKED SOLUTION

The key implication in both parts is

$$AB = 0 \quad \Longrightarrow \quad \operatorname{Im}(B) \subseteq \ker A.$$

Indeed, every $y \in \operatorname{Im} B$ has the form $y = Bx$. Thus $Ay = A(Bx) = (AB)x = 0$, so $y \in \ker A$.

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EXERCISE 8.19 EXAM QUESTION CONTINUED

1. The inclusion and rank-nullity give

$$\begin{aligned}\operatorname{rank}(B) &= \dim \operatorname{Im}(B)\\ &\le \dim \ker A\\ &= n - \operatorname{rank}(A)\\ &= n - (n-1) = 1.\end{aligned}$$

   Since $B \neq 0$, at least one column of B is nonzero, so $\operatorname{rank}(B) \ge 1$. The two bounds yield $\operatorname{rank}(B) = 1$.

2. Apply the same inclusion in $\mathbb{R}^{2031}$:

$$\begin{aligned}\operatorname{rank}(B) &= \dim \operatorname{Im}(B)\\ &\le \dim \ker A\\ &= 2031 - \operatorname{rank}(A).\end{aligned}$$

   Rearranging gives

$$\operatorname{rank}(A) + \operatorname{rank}(B) \le 2031.$$

If every output of B is annihilated by A, the entire image of B must lie in the kernel of A. The dimension of that kernel therefore bounds the rank of B.

You know what that means? V-I-C-T-O-R-Y!

EXERCISE 8.20 CHALLENGE

$$A \in \mathbb{F}^{m\times n}, \quad B \in \mathbb{F}^{n\times p}.$$

$$\operatorname{rank}(AB) = \operatorname{rank} B - \dim(\operatorname{Col} B \cap \ker A).$$

$$A_0 = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B_0 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{over } \mathbb{R}.$$

Please prove the identity and please deduce $\operatorname{rank}(AB) \ge \operatorname{rank} A + \operatorname{rank} B - n$. Please verify the identity for A_0, B_0.

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WORKED SOLUTION

Apply rank-nullity to the restriction L. Its image and kernel are exactly the spaces in the identity. In the example, the kernel of A_0 lies in the column space of B_0, so their intersection has dimension 1.

$$L : \operatorname{Col} B \to \mathbb{F}^m, \qquad L(y) = Ay, \quad \operatorname{Im} L = \operatorname{Col}(AB), \quad \ker L = \operatorname{Col} B \cap \ker A.$$

$$\operatorname{rank} B = \dim \operatorname{Im} L + \dim \ker L, \qquad \dim \ker L \le \dim \ker A = n - \operatorname{rank} A.$$

$$\ker A_0 = \operatorname{span}\{(2,-1,1)^T\}, \quad (2,-1,1)^T = B_0 e_1 - B_0 e_2, \quad \operatorname{rank} B_0 = 2,$$

$$A_0 B_0 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \operatorname{rank}(A_0 B_0) = 1 = 2 - 1.$$

Another one down. Victory!

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